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Layered structure, piezoelectric device using the same, and method of manufacturing piezoelectric device

Abstract

The occurrence of cracking in a functional layer is suppressed, while maintaining flexibility of a layered structure. The layered structure includes a polymer substrate, and a crystalline functional layer formed on the first surface of the substrate. The surface roughness of the first surface of the substrate is 3 nm or less in terms of arithmetic mean roughness (Ra).

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application is a National Stage of International Application No. PCT/JP2020/010177, filed on Mar. 10, 2020, which designates the United States and was published in Japan, and which is based upon and claims priority to Japanese Patent Application No. 2019-052877, filed on Mar. 20, 2019, in the Japan Patent Office. All of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) The present invention relates to a layered structure, a piezoelectric device using the same, and a method of manufacturing a piezoelectric device.

BACKGROUND ART

(3) Conventionally, a piezoelectric device, which makes use of the piezoelectric effect of a substance has been adopted. The piezoelectric effect is a phenomenon in which microscopic polarization occurs in proportion to a pressure applied to a substance. Various sensors such as pressure sensors, acceleration sensors, or acoustic emission (AE) sensors for detecting elastic waves have been manufactured by utilizing the piezoelectric effect.

(4) In recent years, touch panels have been used as input interfaces of information processing devices such as smartphones, and applying piezoelectric devices to touch panels is increasing. A touch panel is integrated into the display device of an information processing device, and high transparency to visible light is required to improve the visibility. On the other hand, it is desirable for the piezoelectric layer to be highly responsive to pressure in order to accurately detect operations by a finger.

(5) A structure in which a piezoelectric layer formed of a metal nitride having a Wurtzite crystal

structure is provided on a flexible insulating film is known (see, for example, Patent Document 1). A structure in which an orientation control layer is inserted between a plastic layer and a piezoelectric layer is also known (see, for example, Patent Document 2).

PRIOR ART DOCUMENTS

(6) Patent Document 1: JP Patent Application Laid-open Publication No. 2018-137268 Patent

Document 2: JP Patent Application Laid-open Publication No. 2018-170503

SUMMARY OF THE INVENTION

Technical Problem to be Solved

(7) When a crystalline functional layer is formed on the surface of a polymer substrate, microcracking often occurs in the functional layer due to the surface roughness of, or foreign substances on the polymer substrate. Even if an electrode film is inserted between the polymer substrate and the crystalline functional layer, it is difficult for the electrode film to absorb the roughness of the substrate surface, and unevenness remains on the surface of the electrode, reflecting the surface roughness of the substrate.

(8) If cracks, pinholes, or the like occur in the functional layer due to the surface roughness of the underlayer, a leakage path which causes electrical short-circuit between the top and bottom electrodes is likely to be produced. This phenomenon becomes more prominent as the functional layer becomes thinner. Examples of the functional layer include a piezoelectric layer, a moisture-sensitive film, an odor-sensitive film, etc. If the functional layer cracks, the performance of the device deteriorates. Upon occurrence of a leakage path, the device may not function any longer.

(9) An objective of the invention is to provide a layered structure capable of suppressing cracks from occurring in a functional layer, while maintaining the flexibility, and to provide a piezoelectric device using such a layered structure and a method of manufacturing the piezoelectric device.

Solution to Solve the Technical Problem

(10) In the first aspect of the invention, a layered structure includes a polymer substrate, and a crystalline functional layer formed on the first surface of the substrate. The surface roughness of the first surface of the substrate is 3 nm or less in terms of arithmetic mean roughness (Ra).

(11) In a second aspect of the invention, a piezoelectric device includes a polymer substrate, a piezoelectric layer formed on the first surface of the substrate, and a pair of electrodes provided on an upper side and a lower side of the piezoelectric layer in the layered direction. The surface roughness of the first surface of the substrate is 3 nm or less in terms of arithmetic mean roughness (Ra).

Advantageous Effect of the Invention

(12) According to the above-described configurations, occurrence of cracks can be suppressed in the functional layer, while maintaining the flexibility of the layered structure.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram of a layered structure according to a first embodiment;

(2) FIG. 2 is a diagram showing surface roughness required for a device characteristic;

(3) FIG. 3 shows an example configuration of a piezoelectric device;

(4) FIG. 4 shows another example configuration of the piezoelectric device;

(5) FIG. 5 shows still another example configuration of the piezoelectric device;

(6) FIG. 6 is a schematic diagram of a layered structure according to a second embodiment;

(7) FIG. 7 shows an example configuration of a piezoelectric device using the layered structure of FIG. 6; and

(8) FIG. 8 shows an example configuration of a piezoelectric device using a layered structure

according to a third embodiment.

DESCRIPTION OF EMBODIMENTS

First Embodiment

(9) FIG. 1 is a schematic diagram of a layered structure **10** according to the first embodiment. The layered structure **10** includes a polymer substrate **11**, and a crystalline functional layer **12** provided on one surface of the substrate **11**. The functional layer **12** is a layer having a specific function, such as a piezoelectric layer in which micro polarization occurs in response to a pressure applied, a temperature-sensitive magnetic film whose hysteresis characteristics change depending on temperature, or a crystalline nanoporous metal oxide film having a catalytic property.

(10) By using the polymer substrate **11**, the entirety of the layered structure **10** can be made flexible. In order to increase the flexibility, it may be desirable to reduce the thickness of the functional layer **12** to such an extent not to damage the expression of the function. However, the thinner the functional layer **12**, the greater the influence of the surface condition of the underlayer, and cracks penetrating through the functional layer **12** may occur.

(11) In the embodiment, the surface roughness of the substrate **11** on which the functional layer **12** is formed is set in a range suitable for exhibiting the device function.

(12) The polymer used for the substrate **11** is, for example, polyethylene terephthalate (PET), polyethylene naphthalate (PEN), polycarbonate (PC), an acrylic resin, a cycloolefin polymer, polyimide (PI), or the like. Among these, polyethylene terephthalate (PET), polyethylene naphthalate (PEN), polycarbonate (PC), acrylic resin, and cycloolefin polymer are colorless and transparent materials.

(13) The thickness of the substrate **11** is, for example, 20 to 200 μm , from the viewpoint of satisfying both flexibility and mechanical strength to support the layered structure.

(14) A piezoelectric layer is formed as the functional layer **12** on the first surface (upper surface in FIG. 1) of the substrate **11**. The piezoelectric layer is formed of, for example, a wurtzite crystal material.

(15) A wurtzite crystal structure is represented by general formula AB, where A is a positive cation ($A.\text{sup.n+}$) and B is a negative anion ($B.\text{sup.n-}$). It is desirable to select a wurtzite piezoelectric material which can express piezoelectric properties to a certain degree or higher, and which can be crystallized in a low temperature process at or below 200° C. For example, zinc oxide (ZnO), zinc sulfide (ZnS), zinc selenide (ZnSe), zinc telluride (ZnTe), aluminum nitride (AlN), gallium nitride (GaN), cadmium selenide (CdSe), cadmium telluride (CdTe), silicon carbide (SiC), etc. can be suitably used. One or a combination of two or more of the above-described compounds may be used.

(16) When two or more of the compounds are combined, layers of the selected compounds may be stacked one by one, or alternatively, a single layer may be formed using the targets of the selected compounds. The selected one or the combination of two or more of the compounds may be used as the main component, and other components may be optionally added. If a component other than the main component is added, the content of the component added as a sub-component is preferably from 0.1 to 30 at. %.

(17) For example, a wurtzite material containing ZnO or AlN as the main component is used. A metal which that does not cause the main component to exhibit conductivity when incorporated in the main component may be added as a dopant. Such dopants may include silicon (Si), magnesium (Mg), vanadium (V), titanium (Ti), zirconium (Zr), lithium (Li), etc. One or a combination of two or more of the above-described dopants may be added. By adding one or more of these metals, the occurrence of cracking can be reduced. When a transparent wurtzite crystalline material is used for the piezoelectric layer, the layered structure is suitable for use in a display.

(18) In order to ensure the flexibility of the layered structure **10** as a whole, it is desirable for the functional layer **12** to be made thin. However, in the case of the piezoelectric layer, it is difficult to achieve a sufficient level of polarization (namely, piezoelectricity) if the film is too thin. For a

functional layer **12** made of a wurtzite piezoelectric material, the film thickness is, for example, 50 nm to 2 μm , preferably 200 nm to 1 μm .

(19) The surface roughness of the first surface of the substrate **11** is 3 nm or less in terms of arithmetic mean roughness (Ra). By using a polymer substrate **11** with a smooth surface, the occurrence of cracking in the functional layer can be suppressed because of the reduced influence from the underlying structure.

(20) FIG. 2 is a diagram showing the range of surface roughness of the substrate **11** suitable for achieving a satisfactory piezoelectric property. The horizontal axis represents arithmetic mean roughness (Ra), and the vertical axis represents full width at half-maximum (FWHM) of the X-ray Rocking Curve (XRC) expressed in degree.

(21) The arithmetic mean roughness (Ra) is a parameter of the height or vertical direction, and represents the average value of the unevenness along the reference length. A part of the roughness curve, which is acquired by measuring the state of the interface using a roughness meter, is extracted along the reference length. In particular, a $1\text{ }\mu\text{m}\times 1\text{ }\mu\text{m}$ surface area is observed by an atomic force microscope (AFM) on various sample substrates. Although an AFM is used to measure the arithmetic mean roughness (Ra) in this example, the measuring method is not limited to the use of AFM. As long as the arithmetic mean roughness (Ra) can be evaluated, any suitable instrument such as a white interferometer or a stylus meter may be used.

(22) XRC FWHM is an index representing crystal orientation. In this example, a ZnO layer having a thickness of 200 nm is formed on each of the sample substrates, and the reflection of the X-rays from the (0002) plane of ZnO is observed by locking curve measurement. The XRC FWHM of the reflected light from the (0002) plane indicates the C-axis orientation of the ZnO crystal. The smaller the XRC FWHM value, the better the crystal orientation and the higher the thickness-direction polarization. It is desirable that the XRC FWHM is less than 15 degrees in order to achieve an acceptable piezoelectric property which allows the device to operate. The surface roughness (Ra) of the substrate **11** corresponding to this XRC FWHM range is 3 nm or less.

(23) In FIG. 2, a data point D with a surface roughness of 5.6 nm and an XRC FWHM of 12.5 degrees is observed. This data point is due to local protrusions present in the area extracted from the sample substrate. Excluding this data point, the arithmetic mean roughness (Ra) of this area is about 1.8 nm, which is within the region A where satisfactory piezoelectric property is achieved.

(24) From this measurement result, it is derived that the surface roughness of the substrate **11** is 3 nm or less in order to achieve satisfactory piezoelectric property of the functional layer **12**. This applies not only to the piezoelectric film, but also to other functional crystal films such as a temperature-sensitive magnetic layer or a catalyst layer, for achieving good crystal orientation with few pinholes or cracks. If these functional layers **12** are formed on the polymer substrate **11**, it is similarly desirable for the surface roughness of the underlying substrate **11** to be 3 nm or less in arithmetic mean roughness.

(25) Examples of the substrate with the above-described range of surface roughness include PET, PEN, PC, acrylic resin, cycloolefin polymer, polyimide, etc. Substrates made of these materials and having undergone a surface smoothing treatment may also be used.

(26) FIG. 3 is a schematic diagram of a piezoelectric device **20A** to which the layered structure **10** is applied. The piezoelectric device **20A** has a layered structure **10a** in which a piezoelectric layer **12a** is provided as the functional layer **12** on the polymer substrate **11**. In the piezoelectric device **20A**, a first electrode layer **13**, the polymer substrate **11**, the piezoelectric layer **12a**, and a second electrode layer **14** are stacked in this order.

(27) When pressure or tensile stress is applied to the piezoelectric device **20A**, polarization proportional to the pressure occurs in the piezoelectric layer **12a**. By extracting the electric charges as an electric current through the first electrode layer **13** and the second electrode layer **14**, the applied pressure or tensile stress can be detected.

(28) Further, by applying a voltage to the piezoelectric layer **12a** via the first electrode layer **13** and

the second electrode layer **14**, an inverse piezoelectric effect that causes the piezoelectric layer **12a** to deform in the layered direction may be utilized. In this case, the piezoelectric device **20A** can be used as a piezo actuator.

(29) The materials of the first electrode layer **13** and the second electrode layer **14** are not limited as long as they are good conductors. When the piezoelectric device **20A** is applied to a light transmissive (optically transparent) device, a transparent electrode may be used for one or both of the first electrode layer **13** and the second electrode layer **14**, depending on the light incident direction and light emitting direction.

(30) The piezoelectric device **20A** is manufactured by, for example, the following process. A polymer film having a surface roughness (arithmetic mean roughness R_a) of 3 nm or less is prepared and used as the substrate **11**. One of the main surfaces of the substrate **11**, on which the piezoelectric layer **12a** is to be formed, is referred to as “the first surface”. The surface roughness (R_a) of the first surface is 3 nm or less.

(31) The first electrode layer **13** is formed on the surface (back surface in FIG. 3) on the opposite side of the substrate **11** from the first surface. For example, an indium tin oxide (ITO) film is formed as the first electrode layer **13** by direct current (DC) or radio frequency (RF) magnetron sputtering.

(32) A ZnO piezoelectric layer **12a** having a film thickness of 50 to 200 nm is formed on the first surface of the substrate **11** by, for example, RF magnetron sputtering using a ZnO target. The material of the piezoelectric layer **12a** is not limited to ZnO, and other materials such as ZnS, ZnSe, ZnTe, AlN, GaN, CdSe, CdTe, SiC, or a combination of two or more of these materials may be used, as described above. Alternatively, one or a combination of two or more of these materials may be used as the main component, and a metal such as Si, Mg, V, Ti, Zr, or Li, which does not cause the piezoelectric layer to exhibit conductivity, may be added.

(33) If a high-dielectric polymer film is used as the substrate **11**, the substrate **11** can be used as a part of the functional layer. If a conductive polymer film is used as the substrate **11**, the substrate **11** can be used as a part of the electrode.

(34) The second electrode layer **14** is formed on the piezoelectric layer **12a**. An ITO film may be formed as the second electrode layer **14** by DC or RF magnetron sputtering, for example.

(35) After the second electrode layer **14** is formed, the ITO films may be crystallized at a temperature below the melting point or the glass transition point (e.g., 130° C.) of the polymer substrate **11** to reduce the electrical resistance of the first electrode layer **13** and the second electrode layer **14**; however, this heat treatment is not essential.

(36) In the piezoelectric device **20A**, the piezoelectric layer **12a** is formed on the first surface of the substrate **11** having an arithmetic mean roughness R_a of 3 nm, and accordingly, cracks are suppressed and good polarization characteristics are achieved. Further, the piezoelectric device **20A** has flexibility as a whole, and the device can be applicable to a wide range of products.

(37) FIG. 4 is a schematic diagram of a piezoelectric device **20B**. In the piezoelectric device **20B**, a polymer film **16** is inserted between the piezoelectric layer **12a** and the second electrode layer **14**. In other words, in the piezoelectric device **20B**, the first electrode layer **13**, the polymer substrate **11**, the piezoelectric layer **12a**, the polymer film **16**, and the second electrode layer **14** are stacked in this order.

(38) As in FIG. 3, the surface roughness of one of the main surfaces (i.e., the first surface) of the substrate **11**, on which the piezoelectric layer **12a** is formed, is 3 nm or less in arithmetic mean roughness (R_a). When the piezoelectric layer **12a** is formed on the first surface of the polymer substrate **11**, cracks can be suppressed, and the polarization characteristics (or piezoelectricity) can be maintained satisfactorily.

(39) The piezoelectric device **20B** is manufactured by, for example, the following process. The piezoelectric layer **12a** is formed on the first surface of the substrate **11**, and the first electrode layer **13** is formed on the surface on the opposite side of the substrate from the first surface, whereby the

first part of the device is prepared. Meanwhile, a second electrode layer **14** is formed on one surface of the polymer film **16** to prepare the second part of the device.

(40) The other surface of the polymer film **16** of the second part is brought so as to face the surface of the piezoelectric layer **12a** of the first part, and bonded to the first part with an adhesive (not shown).

(41) With this configuration, flexibility can be enhanced by providing polymer layers on both sides of the piezoelectric layer **12a** in the layered direction. Further, even if a fine crack occurs in the piezoelectric layer **12a**, occurrence of a leakage path which causes electrical short-circuits between the top and bottom electrodes can be prevented.

(42) FIG. 5 is a schematic diagram of a piezoelectric device **20C**. In the piezoelectric device **20C**, the polymer film **16** is provided on the second electrode layer **14**. That is, the first electrode layer **13**, the polymer substrate **11**, the piezoelectric layer **12a**, the second electrode layer **14**, and the polymer film **16** are stacked in this order.

(43) Similar to FIG. 3 and FIG. 4, the surface roughness of at least one of the main surfaces of the substrate **11**, namely, the surface roughness of the first surface on which the piezoelectric layer **12a** is formed, is 3 nm or less in arithmetic average roughness Ra. By forming the piezoelectric layer **12a** on the first surface of the polymer substrate **11**, cracks can be suppressed, and the polarization characteristics (or piezoelectricity) can be maintained satisfactorily.

(44) The piezoelectric device **20C** is manufactured by, for example, the following process. The piezoelectric layer **12a** is formed on the first surface of the substrate **11**, and the first electrode layer **13** is formed on the surface on the opposite side of the substrate **11** from the first surface to prepare the first part of the device. Meanwhile, a second electrode layer **14** is formed on one surface of the polymer film **16** to prepare the second part of the device.

(45) The second electrode layer **14** of the second part is brought so as to face the surface of the piezoelectric layer **12a** of the first part, and bonded to the first part with an adhesive (not shown).

(46) In this configuration, by arranging the polymer material layers above and below the piezoelectric layer **12a** in the layered direction, the flexibility can be enhanced and the occurrence of cracks in the piezoelectric layer **12a** can be suppressed. The uppermost polymer film **16** can function as a protective film.

Second Embodiment

(47) FIG. 6 is a schematic diagram of a layered structure **30** according to the second embodiment. The layered structure **30** includes a polymer substrate **11**, an amorphous layer **33** provided on the substrate **11**, and a crystalline functional layer **12** provided on the amorphous layer **33**.

(48) The surface roughness of the substrate **11** is 3 nm or less in arithmetic average roughness Ra. By using a polymer base material **11** having such a smooth surface, the entirety of the layered structure **30** can be made flexible and the crystal orientation of the functional layer **12** can be improved.

(49) In the second embodiment, the crystal orientation of the functional layer **12** is further improved by inserting an amorphous layer **33** between the polymer substrate **11** and the functional layer **12**. The “amorphous layer” does not have to be 100% amorphous, and it refers to a layer in which 90% or more, preferably 95% or more is amorphous, and in which the interface with the functional layer **12** is amorphous.

(50) If the layered structure **30** is applied to a device requiring transparency, the amorphous layer **33** may be formed of a transparent metal oxide. The thickness of the amorphous layer **33** is, for example, 3 to 200 nm, preferably 5 to 100 nm. By forming the crystalline functional layer **12** on the amorphous layer **33**, the functional layer **12** can be grown without being affected by the underlying crystal structure. The functional layer **12** with less strain and good crystal orientation has few defects, and occurrence of cracks is suppressed.

(51) If the amorphous layer **33** is an insulating layer, for example, silicon oxide (SiOx), silicon nitride (SiN), aluminum nitride (AlN), aluminum oxide (Al.sub.2O.sub.3), gallium nitride (GaN),

gallium oxide (Ga.sub.2O.sub.3), etc. can be used. Alternatively, ZnO to which Al.sub.2O.sub.3 and SiO.sub.x are added, or GaN, AlN, ZnO or the like to which at least one of Al.sub.2O.sub.3, Ga.sub.2O.sub.3, SiO.sub.x and SiN is added may be used.

(52) If the amorphous layer **33** is formed of an insulating organic film, an acrylic resin, a urethane resin, a melamine resin, an alkyd resin, a siloxane-based polymer, or the like can be used. As the organic substance, a thermosetting resin composed of a mixture of a melamine resin, an alkyd resin and an organic silane condensate may be used.

(53) The amorphous layer **33** may be formed of an organic film to which a conductive dopant is added, and it may be used as an electrode. Alternatively, the amorphous layer **33** may be formed of a conductive metal oxide and used as an electrode.

(54) As the conductive oxide, ITO, indium zinc oxide (IZO), or the like can be used. These materials are also suitable for devices that require light transmissive characteristics. The amorphous layer **33** of the conductive oxide can be formed by, for example, DC or RF magnetron sputtering to a thickness of 5 to 200 nm, more preferably 10 to 100 nm.

(55) When ITO is used, the content of tin (Sn), namely the content ratio Sn/(In+Sn) may be 5 to 15 wt %. At a content ratio within this range, ITO is transparent to visible light, and an amorphous film can be formed by sputtering at room temperature.

(56) When IZO is used, the content of zinc (Zn), namely the content ratio Zn/(In+Zn) may be around 10 wt %. At this content ratio, IZO is also transparent to visible light, and an amorphous film can be formed by sputtering at room temperature.

(57) The amorphous layer **33** formed of the above-described materials has satisfactory surface smoothness, and allows the c-axis of the wurtzite-type material formed on the amorphous layer to be oriented in the growth direction. In addition, the amorphous layer has a high gas barrier property and can reduce the influence of gas derived from the polymer substrate **11** during film formation.

(58) FIG. 7 is a schematic diagram of the piezoelectric device **40** using the layered structure **30a** in which the functional layer **12** is formed as the piezoelectric layer **12a**. In the piezoelectric device **40**, the amorphous layer **33**, the piezoelectric layer **12a**, and the electrode layer **14** are stacked in this order on the polymer substrate **11**. The amorphous layer **33** is formed of a conductive oxide and functions as a first electrode layer.

(59) The piezoelectric device **40** is manufactured by, for example, the following process. A polymer film having a surface roughness (arithmetic mean roughness Ra) of 3 nm or less of the first surface is prepared, which is used as the substrate **11**. An amorphous layer **33** having a thickness of 5 to 100 nm is formed of a conductive oxide such as ITO, IZO, IZTO, or IGZO on the first surface of the substrate **11** by DC or RF magnetron sputtering.

(60) A piezoelectric layer **12a** having a film thickness of 50 to 200 nm is formed on the amorphous layer **33** by, for example, RF magnetron sputtering.

(61) A second electrode layer **14** is formed on the piezoelectric layer **12a**.

(62) In this configuration, the amorphous layer **33** is provided on the smooth polymer substrate **11** having an arithmetic mean roughness Ra of 3 nm or less, and the piezoelectric layer **12a** is provided on the amorphous layer **33**. The crystal orientation of the piezoelectric layer **12a** is further improved, and good piezoelectric characteristics can be obtained.

Third Embodiment

(63) FIG. 8 is a schematic diagram of a layered structure **30b** of the third embodiment and the piezoelectric device **50** using the same. The layered structure **30b** includes a polymer substrate **11**, an amorphous layer **33b** provided on the substrate **11**, a first electrode layer **13** provided on the amorphous layer **33b**, and a piezoelectric layer **12a** provided on the first electrode layer **13**. The piezoelectric layer **12a** is an example of the crystalline functional layer **12**, and has a film thickness of 50 to 200 nm.

(64) The surface roughness of the polymer substrate **11** is 3 nm or less in arithmetic average roughness Ra. By using a polymer substrate **11** with such a smooth surface, the entirety of the

layered structure **30b** can be flexible, and the crystal orientation of the piezoelectric layer **12a** is improved.

(65) In the third embodiment, the amorphous layer **33b** is provided between the substrate **11** and the first electrode layer **13**, which is provided under the piezoelectric layer **12a**. The amorphous layer **33b** is formed of an inorganic material, an organic material, or a mixture of an inorganic material and an organic material. The “amorphous layer” does not have to be 100% amorphous, and it refers to a layer in which 90% or more, preferably 95% or more is amorphous, and in which the interface with the functional layer **12** is amorphous.

(66) By providing the amorphous layer **33** under the first electrode layer **13**, the crystal orientation of the first electrode layer **13** is improved, and the crystal orientation of the piezoelectric layer **12a** formed on the first electrode layer **13** is further improved.

(67) The piezoelectric device **50** can be fabricated by forming the second electrode layer **14** on the top of the layered structure **30b**. Because the piezoelectric device **50** uses the layered structure **30b**, the crystal orientation of the piezoelectric layer **12a** is improved, and the piezoelectric characteristics are improved.

(68) Although specific embodiments of the present invention have been described, the present invention is not limited to the above-described exemplified structures. For example, the layered structure **30** of FIG. **6** may be applied to the device configuration of FIG. **4** or FIG. **5**. By forming the amorphous layer **33** of a conductive organic material, the flexibility of the entire piezoelectric device can be further improved.

(69) The present application is based upon and claims priority to the earlier Japanese Patent Application No. 2019-052877 filed Mar. 20, 2019, the entirety of which is incorporated.

LISTING OF SYMBOLS

(70) **10, 10a, 30, 30a, 30b**: layered structure **11**: substrate **12**: functional layer **12a**: piezoelectric layer **13**: first electrode layer **14**: second electrode layer **16**: polymer film **20A-20C, 40, 50**: piezoelectric device **33**: amorphous layer

Claims

1. A piezoelectric device comprising: a polymer substrate; a piezoelectric layer formed on a first surface of the substrate; a pair of electrodes including an upper electrode and a lower electrode, the upper electrode being provided on an upper side of the piezoelectric layer, and the lower electrode being provided on a lower side of the piezoelectric layer in a layered direction; and an amorphous conductive layer between the substrate and the lower electrode, wherein a surface roughness of the first surface of the substrate is 3 nm or less in arithmetic mean roughness (Ra).
2. The piezoelectric device as claimed in claim 1, wherein, the amorphous conductive layer between the substrate and the lower electrode is a first amorphous conductive layer, and the lower electrode is a second amorphous conductive layer.
3. The piezoelectric device as claimed in claim 2, wherein the second amorphous conductive layer is formed of an oxide conductor or a conductive organic material.
4. The piezoelectric device as claimed in claim 2, wherein the lower electrode is provided between the substrate and the piezoelectric layer.
5. The piezoelectric device as claimed in claim 1, wherein the amorphous conductive layer is formed of an inorganic material, an organic material, or a mixture of an inorganic material and an organic material.
6. The piezoelectric device as claimed in claim 1, wherein a thickness of the substrate is 20 to 200 μm .
7. The piezoelectric device as claimed in claim 1, wherein a thickness of the piezoelectric layer is 50 nm to 2 μm .
8. A method of manufacturing a piezoelectric device, comprising: forming an amorphous layer on a

first surface of a polymer substrate, a surface roughness of the first surface being 3 nm or less in arithmetic mean roughness (Ra); forming a lower electrode on the amorphous layer; forming a piezoelectric layer on the lower electrode; forming the piezoelectric layer on the lower electrode; and forming an upper electrode on the piezoelectric layer.
